

AACHEN, Germany

WMO: 105010

Lat: 50.783N

Lon: 6.095E

Elev: 202

StdP: 98.92

Time Zone: 1.00 (EUC)

Period: 82-03

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.1	-7.1	-14.2	1.1	-7.6	-11.7	1.4	-5.7	13.1	9.1	11.7	7.8	2.0	80	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	28.7	19.2	27.0	18.5	25.3	17.9	20.2	26.6	19.4	25.5	18.6	23.9	2.8	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.3	22.4	17.2	12.6	21.7	16.4	11.9	21.1	58.9	27.2	56.1	25.7	53.3	24.1	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.4	9.1	7.9	DB	-10.2	32.8	4.7	1.7	-13.6	34.0	-16.3	35.0	-18.9	36.0	-22.3	37.2	(4)
(5)			WB	-10.9	22.0	4.5	1.2	-14.1	22.8	-16.7	23.5	-19.2	24.1	-22.5	25.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	2.4	2.2	6.6	8.4	12.9	15.9	18.5	18.2	14.7	11.2	6.8	4.5	(6)	
	DBStd	6.79	4.99	5.36	3.44	3.71	3.83	3.60	3.32	3.08	3.04	3.66	4.02	3.57	(7)	
	HDD10.0	974	238	220	116	73	15	1	0	0	1	29	110	172	(8)	
	HDD18.3	3067	496	451	362	297	174	92	42	41	115	224	346	428	(9)	
	CDD10.0	1064	1	3	12	26	104	177	262	254	142	66	14	3	(10)	
	CDD18.3	114	0	0	0	0	4	19	46	36	7	2	0	0	(11)	
	CDH23.3	843	0	0	0	0	47	129	362	255	46	4	0	0	(12)	
	CDH26.7	180	0	0	0	0	6	13	88	68	5	0	0	0	(13)	
(14) Wind	WSAvg	3.1	4.3	3.6	3.8	3.3	2.5	2.5	2.2	2.4	2.8	3.0	3.6	4.0	(14)	
(15) Precipitation	PrecAvg	814	68	65	55	48	62	77	79	84	64	64	69	79	(15)	
	PrecMax	987	137	153	114	97	130	160	186	170	158	152	115	174	(16)	
	PrecMin	652	5	11	5	1	22	19	30	25	16	33	4	22	(17)	
	PrecStd	93	33	32	28	23	25	33	40	42	38	25	28	34	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.0	16.1	19.6	21.5	27.1	28.3	31.2	31.7	27.4	23.8	17.1	14.6	(19)	
		MCWB	9.0	10.6	11.2	12.2	16.3	19.3	20.0	20.0	17.7	17.2	11.9	9.9	(20)	
	2%	DB	10.6	13.0	16.4	19.0	24.6	26.5	29.0	28.4	24.5	20.4	15.1	12.9	(21)	
		MCWB	7.9	9.0	10.1	11.0	15.6	18.5	19.6	19.1	17.4	15.6	11.1	9.7	(22)	
	5%	DB	9.4	11.1	13.7	16.7	22.1	24.9	27.3	26.1	21.8	18.3	13.5	10.9	(23)	
		MCWB	7.6	8.1	9.7	10.3	14.4	17.8	19.0	18.4	16.0	14.4	10.7	8.7	(24)	
	10%	DB	8.2	9.3	11.9	14.7	19.9	22.6	25.3	24.0	19.7	16.3	12.2	9.3	(25)	
		MCWB	6.5	7.1	8.7	9.4	13.7	16.7	18.4	17.6	15.1	13.2	10.1	7.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.3	11.4	12.1	13.4	17.5	20.7	21.5	21.2	19.0	17.5	13.0	11.1	(27)	
		MCDB	11.6	14.7	17.5	19.1	25.0	25.9	28.0	27.7	25.6	22.2	15.0	12.9	(28)	
	2%	WB	8.8	9.9	11.0	11.8	16.3	19.5	20.5	19.9	17.6	16.1	12.0	10.1	(29)	
		MCDB	9.8	12.5	14.9	17.1	22.7	25.0	27.1	26.5	22.6	19.7	13.9	12.4	(30)	
	5%	WB	7.8	8.4	10.1	10.9	15.1	18.5	19.7	19.0	16.7	14.9	11.1	9.0	(31)	
		MCDB	9.0	10.7	13.5	15.9	20.7	23.5	25.9	24.8	20.5	17.8	13.3	10.6	(32)	
	10%	WB	6.7	7.1	8.9	9.8	14.0	17.2	18.7	18.2	15.8	13.4	10.1	7.9	(33)	
		MCDB	8.2	9.2	11.4	14.1	19.1	21.6	24.2	23.1	19.2	15.7	12.1	9.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.3	5.8	6.4	7.7	8.2	8.1	8.9	8.5	7.5	6.1	4.8	4.0	(35)	
		MCDBR	5.6	7.9	9.1	11.2	12.4	11.6	12.2	12.0	11.1	8.2	6.6	5.7	(36)	
	5% WB	MCWBR	4.6	5.3	5.4	5.8	6.1	5.8	5.6	5.3	5.5	4.7	4.5	4.3	(37)	
		MCWBR	4.8	7.1	7.7	9.9	11.1	10.6	11.1	10.1	9.4	7.3	5.8	5.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.314	0.337	0.369	0.403	0.410	0.412	0.415	0.407	0.378	0.354	0.330	0.310	(40)	
		taud	2.429	2.379	2.306	2.236	2.255	2.266	2.256	2.303	2.370	2.433	2.429	2.402	(41)	
	Ebn at Noon		717	791	826	835	846	846	836	823	808	760	686	657	(42)	
		Edh at Noon	61	82	105	126	129	129	128	117	98	77	61	54	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.76	1.44	2.58	4.11	4.88	5.33	5.05	4.32	3.21	1.89	0.89	0.59	(44)	
	RadStd		0.11	0.28	0.38	0.47	0.46	0.49	0.61	0.43	0.37	0.17	0.11	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (84 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page